Curriculum Vitae

Naseem Ahamad, Ph.D.
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Objective

I am a highly motivated and accomplished researcher with a deep passion for stem cell and molecular biology, calcium signaling, autoimmune diseases such as Sjögren's syndrome, immunology, and infectious disease research. My strong track record includes comprehensive investigations into stem cell dynamics, immune responses in various disease conditions, and the complexities of viral and bacterial infections. I am eager to apply my diverse skill set and enthusiasm for translational research to make meaningful contributions to your ongoing research, expand my knowledge in a collaborative, and contribute to achieving your dynamic research goal.

Publications

- **Ahamad N*, Chaudhary JK (2025). Interferon Regulatory Factors in Human Diseases: Guardians of Cellular Harmony and Their Role in Pathological Imbalance.** Under preparation. *** Corresponding author**
- Chaudhary J.K., **Ahamad N**., Rath P.C. **(2024) Mesenchymal stem cells (MSCs) from the mouse bone marrow show differential spatio-temporal expression of Interferon Regulatory Factors (IRF-1 and IRF-2)** Mol Biol Rep. 2024 Jan 9;51(1):97. doi: 10.1007/s11033-023-09025-9. **PMID: 38194130**
- **Ahamad, N***. Gupta, S., Parashar, D. **(2022)**. **Using Omics to Study Leprosy, Tuberculosis, and Other Mycobacterial Diseases**. Front Cell Infect Microbiol 2022 Feb 24;12:792617. doi: 10.3389/fcimb.2022.792617. **PMID: 35281437 * Corresponding author**
- **Ahamad, N**. Sun, Y., Da Conceicao, RD Caroline Regina Delma, Natarajan M, Singh, B. B. **(2021)**. **Differential activation of Ca2+ influx channels modulate stem cell potency, their proliferation/viability and tissue regeneration**. NPJ Regen Med 2021 Oct 20;6(1):67. **PMID: 34671058**
- **Ahamad, N**. Sun, Y, Singh B. B. **(2021)**. **Increasing cytosolic Ca2+ levels restore cell proliferation and stem cell potency in aged MSCs**. Stem Cell Res 2021 Oct;56:102560 **PMID: 34624617**
- Sukumaran P., Sun, Y., Da Conceicao, Selvaraj S, **Ahamad N**., Singh, B. B. **(2021)**. **Calcium Signaling Regulates Autophagy and Apoptosis.** Cells 2021 Aug 18;10(8):2125. **PMID: 34440894**
- **Ahamad, N**., Singh, B. B. **(2021)**. **Calcium channels and their role in regenerative medicine**. World J Stem Cells 2021 April 26; 13(4): 260-280 **PMID: 33959218**
- Sun, Y., Da Conceicao, V. N., **Ahamad, N.**, Madesh, M., Singh, B. B. **(2020)**. **Spatial localization of SOCE channels and its modulators regulate neuronal physiology and contributes to pathology**. Current Opinion in Physiology. 2020 Jul; 17, 50-62, https://doi.org/10.1016/j.cophys.2020.07.008.
- **Ahamad N**., Rath P.C. **(2019) Expression of interferon regulatory factors (IRF-1 and IRF-2) during radiation-induced damage and regeneration of bone marrow by transplantation in mouse.** Mol Biol Rep. 2019 Feb; 46(1):551-567. doi: 10.1007/s11033-018-4508-x. **PMID: 30488374**.
- Anand BG, Prajapati KP, Dubey K, **Ahamad N**, Shekhawat DS, Rath PC, Joseph GK, Kar K. **(2019) Self- Assembly of Artificial Sweetener Aspartame Yields Amyloid-Like Cytotoxic Nanostructures**. ACS Nano. 2019 Apr 25. doi: 10.1021/acsnano.9b02284. **PMID:31021591**
- **Ahamad N.** and Rath P. C**. (2019) Bone Marrow stem cells, Aging and Age-related Diseases. In Models, Molecules and Mechanisms in Biogerontology** (pp. 321-352). Springer Nature, Singapore (https://www.springer.com/us/book/9789811335846).

Experience

- **Post-Doctoral Fellow (**2019 - 2024), University of Texas Health Science Center, San Antonio, USA. I worked, applied my Ph.D. experience, and **earned expertise in stem cells, calcium, calcium channels, signaling, and salivary gland regeneration**.

Education

- **Ph.D. (**Dec **2017)** specializing in **Stem Cell Biology and Molecular Biology,** Jawaharlal Nehru University (JNU), New Delhi. Title of Ph.D. Thesis: "**Bone marrow regeneration in the mouse using bone marrow-derived stem cells.**"

- **M.Sc. (**Jul **2009**, 70.50%**)** specializing in **Microbiology** Gurukul Kangri Vishvavidyalya, Hardwar, UK.
  Title of M.Sc. Thesis: "**Detection of single nucleotide polymorphism in TLR-9 gene of T.B. patient and their healthy contacts by BiPASA technique**."

Specific skills/expertise

- **Stem Cells and Mammalian Cells culture :** Isolation, culture, and propagation of mouse **bone marrow-derived mesenchymal stem cells (mMSCs)** and **human salivary gland-derived stem cells (hSGSCs)**. Experimentation on cell lines such as THP-1, HEK-293, HeLa, 293FT and S.H.- SY5Y Cell Line (**Transfection**, **MTT assay** for cell viability, growth, and **BrdU** for proliferation). **Colony-forming Unit-Fibroblast (CFU-Fs) assay** following **Crystal Violet (CV) staining** and **Giemsa staining** to enumerate stem cell potential in the total population. Study the Effect of γ-radiation on the colony-forming potential of the whole bone marrow. **Isolation of bone marrow stem cells** from bone marrow tissue and **transplantation to regenerate the irradiation-damaged bone marrow**. **Salivary gland stem cells** are isolated from salivary gland tissue and culture for transplantation to regenerate the damaged gland in mouse model. **Isolation of lung stem cells (AT2)** from lungs and **generation of the organoid model**. Study stem cell aging using calcium. Isolation of bone marrow stem cells from bone marrow tissue and transplantation to repopulate the bone marrow in a chemically induced neurodegenerative mouse model.
- **Characterization of stem cells by FACS: Characterized Bone marrow** and **mesenchymal stem cells** by **Fluorescence-Activated Cell Sorter (FACS)** using an antibody against their cell surface markers (MSCs positive CD29, CD44, Sca-1and MSC negative markers CD11b, CD34, CD45). This technique has been instrumental in my research, allowing me to conduct **Cell cycle analysis** of bone marrow and MSCs and study the Effect of γ-radiation on the DNA content of total bone marrow. Used FACS to **analyze the calcium levels of different passage MSCs.** Investigate **MSC differentiation into osteocytes** and **adipocytes**. **Characterized Hematopoietic stem cell lineages** such as LT-HSCs, ST-HSCs, LSK, MPP3, MPP4, GMPs and CMPs after pneumonia infection by Fluorescence-Activated Cell Sorter (FACS). Analyzed the engraftment potential of transplanted stem cells using FACS..
- **Microscopy techniques:** Conducted **Histological study (focus score)** using Bright/Phase light microscopy, Differential interference contrast (DIC), and **Stereomicroscopy**. Studied the expression level of surface and intracellular protein by **immunofluorescence**. Familiar with **Confocal/Live cell imaging** and **calcium imaging techniques**.
- **Microbial techniques: Cultured and isolated bacteria using the Membrane filtration technique**. Conducted the **MPN Method for E. coli**, **Gram staining**, and microscopic observations. **Experienced in Media formulation** (L.J. Media). Performed Drug Sensitivity Test. **Conducted Transfection assay**. **Culture and bacterial Kpn infection** in mice for disease to study lung pathology.
- **Molecular biological techniques: Isolated RNA and DNA from different types of cells and tissues**. **Quantified RNA and DNA** using nanodrop, performed **Real-Time RT-PCR** to study the expression level. Performed **protein quantification using Bradford's method**, **SDS-PAGE, Western Blotting (DAB), CCB, and Silver staining**. Experienced in cDNA preparation and primer designing and conducted **gene silencing by siRNA**.
- **Animal Breeding:** Six or more years of **experience in breeding, handling, and maintaining mice**, including wild-type, **transgenic, and knock-out strains**. Experience in maintaining records and pedigrees for mouse colonies. Skilled in proper handling and care of mice, including feeding and watering. Knowledge of standard procedures for **mouse genotyping** and phenotyping. Familiar with the use of various mouse models in research such as FAD and ALS mouse model.
- **Animal Bone Marrow Transplantation technique:** Experience in **transplanting mice's bone marrow cells and mesenchymal stem cells after radiation**. Studies conducted on **bone marrow and salivary gland regeneration** using this method. Isolation and culture of salivary gland stem cells for transplantation to regenerate damaged glands. Experience in **creating 3D organoid models for lung studies**. Isolation and **transplantation of CD8+ T cells and giving retro-orbital into a disease model**. Isolation and **transplantation of BMCs into neurodegenerative disease models**.

Scientific lectures and talks

- **Ahamad N*** (**Jan 21, 2022**). **"Tissue regeneration in the mouse by using bone marrow-derived stem cells."** at Sanford Research, Sioux Falls SD 57104, USA.
- **Ahamad N*** and Rath P.C. (**Nov 2016**). **"Characterization of bone marrow and mesenchymal stem cells for the bone marrow transplantation in the mouse"** in **Indian Ageing Congress (IAC) 2016,** 18th Biennial Conference of Association of Gerontology (India) & 14th Annual Conference of Indian Academy of Geriatrics on **Nov 9-11, 2016,** at Anatomy Lecture Theater, All India Institute of Medical Sciences (AIIMS), New Delhi, India.

Scientific activities and poster presentations

- **Ahamad N***, Minyoung K Jeong, Joon Paek, Rehae C Miller, Saja Alramadhan, Indraneel Bhattacharrya, and Seunghee Cha. (**Mar 2022**). "**Anti-P2X7 receptor therapy in the Sjögren's syndrome-prone mouse at disease onset ameliorates disease phenotype by reshaping germinal center T cells**" in "**Spring Synergy 2022**", On **Apr 6, 2022,** organized by the College of Dentistry, University of Florida, Gainesville 32608, Florida, USA

- **Ahamad N*** and Rath P.C. (**Mar 2017**). "**Bone marrow regeneration by transplantation of marrow and mesenchymal stem cells in the mouse**" in "**BIOSPARKS 2017**", 15th Annual Scientific Festival on **Mar 30-31, 2017,** organized by the School of Life Sciences, at the auditorium, Jawaharlal Nehru University, (JNU), New Delhi, India.
- **Ahamad N*** and Rath P.C. (**Feb 2017**). **"Bone marrow regeneration by transplantation of bone marrow cells and mesenchymal stem cells in mouse"** in **International Conference on "Mitochondria in Health and Disease",** 6th Annual Conference of Society for Mitochondrial Research and Medicine-India on **Feb 10-11, 2017,** organized by the School of Life Sciences, at Convention Centre, Jawaharlal Nehru University, (JNU), New Delhi, India.

International conferences, meetings, and symposiums attended

- **International Seminar:** Participated/Attended in one day International Seminar on **"New and Innovative Library & Information Services"** organized by **Central Library, Jawaharlal Nehru University, New Delhi, India** in collaboration with International Federation of Library Associations and Institutions **(IFLA)-** Regional Standing Committee for Asia and Oceania **(RSCAO)** and Association for Media Libraries and Archives **(AMLA)** held at Convention Center, JNU, New Delhi, India (22nd **Feb 2017**).
- **Conference:** Participated/Attended in the Indo-German workshop on **"Regenerative Medicine in Liver Diseases"** held at Institute of Liver and Biliary Sciences (ILBS), New Delhi, India (14th Feb to 16th **Feb 2013**).
- **Conference:** Participated/Attended in the **"3rd International Conference on Stem Cell and Cancer (ICSCC-2012): Proliferation, Differentiation and Apoptosis"** held at PGIMER Auditorium, Dr. RML Hospital, New Delhi, India (27th Oct to 30th **Oct 2012**).

National conferences, meetings, and symposiums attended

- **National Science Day Symposium 2017:** Participated in **National Science Day Symposium 2017,** held at on **28 Feb 2017** organized by the School of Life Sciences, at Convention Centre, Jawaharlal Nehru University, (JNU), New Delhi, India. (28th **Feb 2017**).
- **Workshop:** Successfully completed a workshop on **"Combined Confocal Raman-AFM System: Basics & Research Applications"** held at AIRF, Jawaharlal Nehru University, (JNU), New Delhi, India. (2nd February to 3rd **Feb 2017**).
- **Workshop:** Successfully completed a workshop on **"Biological Sample Preparation for Transmission and Scanning Electron Microscopy"** held at AIRF, Jawaharlal Nehru University, (JNU), New Delhi, India. (24th January to 25th **Jan 2017**).
- **Conference:** Participated in **"Current Trends in Hypoxia Research 2017 conference"** held at School of Life Sciences (SLS), Jawaharlal Nehru University, (JNU), New Delhi, India. (18th **Jan 2017**).
- **Annual Open Day 2016:** Participated in **"JNU Annual Open Day 2016"** poster presentation held at Convention Centre, Jawaharlal Nehru University, (JNU), New Delhi, India. (21st **Nov 2016**).
- **Training:** Successfully completed **"Orientation Program and obtained Hands-on Training on Laboratory Animals Mice, Rat and Rabbit for Handling, Care and Procedure"** sponsored by Central Laboratory Animal Resources (CLAR), Jawaharlal Nehru University, (JNU), New Delhi, India. (12th Aug to 13th **Aug 2016**).
- **Workshop:** Successfully completed a workshop on **"Flow Cytometry and its Applications"** held at Jawaharlal Nehru University, (JNU), New Delhi, India. (12th Apr to 13th **Apr 2016**).
- **Workshop:** Successfully completed a workshop on **"Flow Cytometry Experimental Designing and Data Analysis"** held at Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore. India. (11th Mar to 13th **Mar 2015**).
- **Training:** Successfully completed training on **"Safe handling of radioactive sources in research applications"** at Jawaharlal Nehru University, (JNU), New Delhi, India. (05th Aug to 08th **Aug 2014**).

Short-term courses /diploma/ workshop

- **Workshop:** Successfully completed a Two-day workshop on **"Bridges in Communication: Indian Sign Language for Linguistic Capacity in our Community"** organized by the Linguistic Empowerment Cell, Jawaharlal Nehru University, (JNU), New Delhi, India. (20th January to 21st **Jan 2017**).
- **Workshop:** Successfully completed a Two-day workshop on **"EAW: spoken English and Presentation Skills"** organized by the Linguistic Empowerment Cell, Jawaharlal Nehru University, (JNU), New Delhi, India. (4th Dec to 13th **Dec 2015**).
- **Training/Workshop:** Successfully completed **"Capacity Building Program"** sponsored by University Grant Commission (UGC) on **Research and Data Analysis** (Methodology, Experimental designing and writing project report and application of various statistical tools, including SPSS, Sigma plot, Microsoft XL to analyze various kind of data) imparted by Sambodhi Research and Management Institute at Centre for Study of Law and Governance (CSLG), Jawaharlal Nehru University (JNU), New Delhi, India. (Training/Workshop duration 29th Mar to 10th **May 2012**).

Soft Skills

I am a collaborative team player and an independent researcher with strong problem-solving skills. I pay great attention to detail and possess critical thinking abilities. My extensive experience in designing and executing experiments and data analysis skills instill confidence in my capabilities. I have presented my findings at lab meetings and national and international conferences, and I actively share knowledge within the scientific community.

Technical Skills

- **Computer Applications:** Advanced proficiency in the **Microsoft Office Suite (Word, PowerPoint, Excel**). Completed a comprehensive 3-month Basic Computer Course covering MS Office, Windows operating system, and Internet fundamentals. Knowledge of **Power BI, Tableau**.
- **Research Software Applications:** Proficient in laboratory imaging software such as **Alpha Imager, ImageJ, and Adobe Photoshop**. Experienced in **microscopy analysis software** for cellular parameter measurements. Skilled in **flow cytometry** analysis using **Modfit** (for cell cycle analysis) and **FlowJo** (for immunophenotyping). Proficient with **EndNote** for scientific reference management.
- **Statistical Analysis Capabilities:** Strong foundation in descriptive statistics (mean, median, mode). Experienced in calculating measures of variance (standard deviation, standard error). Proficient in hypothesis testing using chi-square tests and Student's t-test. Capable of performing ANOVA and calculating p-values.

Awards and Honors

- **Post-doctoral Fellow**: Nov 2019 – Nov 2024
- **DBT-Research Associate Fellowship**: Jul 2019 – Nov 2019
- **Best Poster Award** BIOSPARKS 2017
- UGC- **Senior Research Fellowship** (SRF): Jul 2013-Jul 2016
- UGC- **Junior Research Fellowship** (JRF): Jul 2011-Jul 2013
- **GATE- 2009, GATE- 2010, GATE- 2011**

Languages

English, Hindi.

Declaration

At this moment, I declare that the above-given information is accurate and correct to the best of my knowledge. I assure you that I will render my duties to the best and satisfaction of my superior.

Date: Fab, 2026

Place: New Delhi

Naseem Ahamad